MODULE 19 · DOMAIN 2 - ML MODEL AND FM DEVELOPMENT

Explainability and experiments

SHAP, partial dependence, post-training bias, and keeping track of what you tried.

MODULE 19 of 43 DOMAIN 2 - Model Dev WEIGHT 24%

admin@sthithapragna.com

A metric says **how well**. Explainability says **why**. Regulators ask the second one.

admin@sthithapragna.com

Four audiences, four different explanations

- **The regulator:** which factors drove this individual decision, and by how much?
- **The customer:** why was I declined, in language I can act on?
- **The data scientist:** is the model using the features I expect, for sensible reasons?
- **The business:** what generally drives this outcome across the whole population?
- The first two need **per-prediction** explanations. The last needs a **global** one.

> ON THE EXAM
>
> Local and global explanations answer different questions. Offering the wrong one is a common and unhelpful mistake.

Clarify is closed to new customers

- **Closed to new customers from 30 July 2026.** Existing customers continue as normal.
- It remains named in the exam guide in Tasks 1.3, 2.3 and 4.1, so it remains **examinable**.
- **The replacement path is documented and good.** The mathematics and the engine are both public.
- Pre-training metrics are published formulas over label counts. Post-training metrics use confusion-matrix counts.
- **Feature attribution uses the SHAP library**, which is the same engine Clarify is built on.

> ON THE EXAM
>
> AWS documents computing these directly with pandas, scikit-learn and SHAP, run in Pipelines and logged to a SageMaker AI MLflow App.

SHAP, and Shapley values

- Comes from **cooperative game theory**: how do you fairly divide credit among contributors?
- Applied to a prediction, it assigns each feature an **importance value for that prediction**.
- **Model-agnostic.** It works on any model, because it only needs to call it repeatedly.
- Values are signed: a feature can push the prediction **up or down**.
- It is expensive, because computing it means many model invocations per explanation.

ON THE EXAM

The signed value is what makes it usable for a regulator: 'income raised the score by 0.12, recent arrears lowered it by 0.31'.

Local or global explanation?

LOCAL

One prediction

- Why was this application declined?
- SHAP values for that single instance.
- What a regulator and a customer need.
- Computed per prediction, so it costs per prediction.

GLOBAL

The whole model

- What drives this outcome generally?
- SHAP averaged across many examples.
- What the business and the data scientist need.
- Computed once over a sample.

THE TIEBREAKER

Global importance is usually the mean absolute SHAP value across examples. Same technique, aggregated.

Explanations are relative to a baseline

- SHAP explains a prediction **against a baseline**, not in absolute terms.
- The baseline represents an **uninformative or average example** from the dataset.
- So 'this feature contributed +0.2' means **relative to that baseline**.
- A badly chosen baseline produces explanations that are technically correct and misleading.
- For images the baseline is often implicit, such as a uniformly coloured image.

> ON THE EXAM
>
> Why was X rather than Y? SHAP answers that, and Y is the baseline. Changing Y changes the explanation.

Partial dependence plots

- Shows how the predicted target changes as you **vary one feature**, holding others as they are.
- It is a **global** view: the marginal effect of a feature across the population.
- Excellent for showing non-linearity, such as risk rising sharply after a threshold.
- It does **not** explain an individual prediction. That is what SHAP is for.
- It can mislead when features are strongly correlated, from module 05.

> ON THE EXAM
>
> A PDP answers 'what does this feature do to the model'. SHAP answers 'what did this feature do to this decision'.

Online explainability

- Returns a **per-instance explanation in real time**, from a SageMaker AI endpoint.
- So the explanation arrives **with the prediction**, rather than in a batch report afterwards.
- That is what a system needs when it must tell a customer why, at the moment of the decision.
- It costs latency and compute on every request, because SHAP needs many model invocations.
- The alternative is explaining a sample after the fact, which suits audit but not a live decision.

> ON THE EXAM
>
> If the explanation must appear in the same response as the decision, that is online explainability and it has a latency price.

Post-training bias metrics

- Module 08 measured bias in the **data**. These measure bias in the **predictions**.
- They are standardised functions of **per-group confusion-matrix counts**.
- They require the **model or endpoint name** in the configuration, because they need predictions.
- So they are **considerably slower** than pre-training metrics, including 3 to 5 minutes of endpoint spin-up.
- A dataset with no measurable pre-training bias can still produce a biased model, via proxies.

> ON THE EXAM
>
> Pre-training bias is arithmetic on labels. Post-training bias needs the model to make predictions first, which is why it costs more.

Post-training bias metrics worth recognising

METRIC	WHAT IT COMPARES
DPPL	Difference in positive proportions in predicted labels
DI	Disparate impact: the ratio of positive prediction rates
RD	Recall difference between the two facets
DAR / DRR	Difference in acceptance rates, and in rejection rates
AD	Accuracy difference between the facets
FT	Counterfactual fliptest: do similar examples across facets get similar predictions?

ON THE EXAM

DPPL is the post-training twin of module 08's DPL: the same comparison, made on predictions instead of labels.

What Clarify does, in one table

CAPABILITY	STAGE AND SCOPE
Pre-training bias metrics	Before training. Data and labels only. Module 08
Post-training bias metrics	After training. Needs model predictions
SHAP feature attribution	After training. Local per instance, or global averaged
Partial dependence plots	After training. Global, one feature at a time
Online explainability	At inference. Per instance, in real time
Bias and feature attribution drift	In production, via Model Monitor. Module 29

ON THE EXAM

Six capabilities across four stages. Questions usually turn on which stage, and whether the answer needs one prediction or the whole model.

You will run **dozens** of experiments. Six months later somebody will ask which one you shipped.

What an experiment record has to hold

- **Parameters:** hyperparameters, feature set, algorithm, container image version.
- **Metrics:** training and validation metrics, per epoch and final.
- **Artefacts:** the model itself, plots, the evaluation report from module 17.
- **Data version:** which dataset produced this. Module 06's point, and the one most often missed.
- Anything not recorded automatically will eventually not be recorded.

> ON THE EXAM
>
> Manual tracking in a spreadsheet works until the week somebody is busy, which is the week the important run happens.

SageMaker managed MLflow

- Fully managed **MLflow tracking servers**, generally available since June 2024.
- Familiar open-source MLflow constructs, with setup in a couple of clicks and **IAM-secured access**.
- The SDK tracks from **local notebooks, IDEs, Studio, Training jobs, Processing jobs and Pipelines**.
- **Models registered in MLflow appear automatically in the SageMaker Model Registry.**
- MLflow models deploy to SageMaker Inference **without custom containers**.

> ON THE EXAM
>
> The Model Registry integration is what makes it more than a notebook tool: experiments and governance share one lineage.

What managed MLflow costs, and where it runs

- **The tracking server is not serverless.** It is an always-on instance, billed continuously.
- So an idle tracking server costs money, in the same way an idle endpoint does.
- Available in all Regions where SageMaker AI is available, **except China and GovCloud (US)**.
- **Migration tooling** exists for moving a self-managed MLflow server from EC2 or on-premises.
- Size the server to the team. It is shared infrastructure, not per-project.

> ON THE EXAM
>
> The same cost shape as an endpoint: provisioned, always on, and easy to forget. Module 32 comes back to this pattern.

SageMaker Experiments

- The earlier SageMaker-native experiment tracking: experiments, trials and trial components.
- Automatically captured metadata from training jobs without explicit instrumentation.
- **Managed MLflow is where AWS is investing now**, and it is the open-source standard.
- The vocabulary still appears in older material and in questions. Recognise it.
- The concept is identical either way: record what you ran, so you can compare and reproduce it.

ON THE EXAM

Two names for the same idea. If a question describes tracking runs and comparing them, either answer is describing the same capability.

What MLflow gives you beyond a record

- **Rich visualisations for run comparison**, so you see the shape of the search rather than a table.
- Plot a metric against a hyperparameter across dozens of runs, and the pattern becomes obvious.
- That closes module 15's loop: tuning results become something you can look at and reason about.
- Runs are grouped into experiments, so one project's history stays together.
- Artefacts are stored alongside, so a plot or evaluation report travels with its run.

> ON THE EXAM
>
> A table of fifty runs is data. A chart of fifty runs is a finding. The visualisation is doing real work.

Lineage

- **Lineage** links a model back through its training job to the data that produced it.
- The question it answers: **which dataset version produced the model now in production?**
- SageMaker AI records much of this automatically as jobs run.
- Feature Store lineage from module 06 connects feature groups to the models trained on them.
- An incident review needs this chain. Reconstructing it afterwards is usually impossible.

> ON THE EXAM
>
> Experiment tracking records what you tried. Lineage records what produced what. An audit needs the second one.

Model cards

- Standardised documentation for a model: **intended use, risk rating, training details, evaluation results**.
- **Integrated with the Model Registry**, and attachable to a registered model version.
- Can include **bias reports and explainability output**, so the governance story sits in one artefact.
- Exportable as PDF for compliance, audit and regulatory reporting.
- Module 27 covers governance properly. This is where the content for it comes from.

> ON THE EXAM
>
> Everything in this module and the last two ends up here: metrics, bias reports, explanations and intended use, in one document.

Debugger and Profiler, briefly

- **Debugger is closed to new customers. Profiler is in the sunset list**, with a termination date.
- Debugger watched tensors during training and raised rules for vanishing gradients, overfitting, poor GPU use.
- Both remain named in the exam guide, so both remain examinable.
- In practice: CloudWatch metrics, your own logging, and MLflow for the run history.
- Module 14 covered this properly. It appears here because it belongs to the same toolkit.

> ON THE EXAM
>
> Debugger diagnosed a run in flight. MLflow records runs after the fact. Neither replaces the other; only one is still available.

Explainability does not stop at deployment

- **Bias drift** watches fairness metrics change over time in production.
- **Feature attribution drift** watches which features the model is actually relying on.
- A shift in feature importance is an **early warning**: the model's reasoning changed before its accuracy did.
- Both run through Model Monitor, which is closed to new customers. Module 29.
- The replacement path logs the same computations to MLflow on a schedule.

> ON THE EXAM
>
> Accuracy drops after the world changes. Feature importance often shifts first, which makes attribution drift the earlier signal.

What explainability costs

ACTIVITY	COST SHAPE
Global SHAP over a sample	A processing job. Bounded, and run occasionally
Online explainability	Added latency and compute on every single request
Post-training bias	A processing job plus model inference, including endpoint spin-up
MLflow tracking server	Always-on instance. Billed whether anyone is running experiments
Drift monitoring	Scheduled jobs, repeated forever. Module 29

ON THE EXAM

Only one row scales with traffic. Online explainability is the expensive one, and it is the one regulation sometimes requires.

Symptoms, and what they usually mean

SYMPTOM	USUAL CAUSE
One feature dominates every explanation	Leakage. Module 05's check, done properly
Cannot reproduce last quarter's best model	No experiment tracking, or no data version recorded
Explanation contradicts domain expertise	Correlated features, or a badly chosen baseline
PDP offered for an individual decision	Wrong scope. PDP is global; SHAP is local
Endpoint latency rose after adding explanations	Online explainability. SHAP costs invocations
Post-training bias job far slower than expected	It needs predictions, plus endpoint spin-up time
MLflow bill with no experiments running	Tracking server is always on, not serverless

ON THE EXAM

Row one is the highest-value check in the whole course: explainability is how you catch a leak before production does.

If you cannot say **why** a model decided, you cannot defend it, fix it, or improve it.

> An insurer must tell each declined applicant, in the decline letter generated at the moment of the decision, which factors most influenced the outcome and in which direction.
>
> The model is an XGBoost classifier on a SageMaker AI real-time endpoint. Batch reports produced overnight are not acceptable, because the letter is issued immediately.

What should they implement?

A. A partial dependence plot for each feature, included in the letter

B. Online explainability, returning per-instance SHAP values with the prediction

C. A scheduled Clarify processing job producing global feature importance

D. Post-training bias metrics computed nightly against the endpoint

◆ PAUSE AND DECIDE

What should they implement?

A. A partial dependence plot for each feature, included in the letter

A PDP is a global view of a feature's marginal effect across the population. It says nothing about this applicant's decision.

B. Online explainability, returning per-instance SHAP values with the prediction

Per-instance, real time, signed values showing direction and magnitude. It arrives with the prediction, which the immediate letter requires.

C. A scheduled Clarify processing job producing global feature importance

Global scope and batch timing. Two mismatches: the letter is about one applicant, and it is issued immediately.

D. Post-training bias metrics computed nightly against the endpoint

Measures fairness across groups, which is a different question entirely. It does not explain an individual decision.

THE SIGNAL

Two constraints, both stated: per-applicant, and at the moment of decision. Only online explainability satisfies both.

A team in a newly created AWS account needs pre-training bias metrics and per-prediction feature attributions for a credit model, to satisfy an internal model risk policy.

They find they cannot create SageMaker Clarify jobs. Their compliance function requires the results to be versioned and auditable alongside the training runs that produced them.

What should they do?

A Request that AWS Support enable Clarify for the account

B Abandon the attribution requirement and report accuracy metrics only

C Use a foundation model to write explanations of each declined application

D Compute the metrics directly with pandas and SHAP in Pipelines, logging to a SageMaker AI MLflow App

◆ PAUSE AND DECIDE

What should they do?

A Request that AWS Support enable Clarify for the account

Closed to new customers is a service lifecycle decision, not a quota. Support cannot reopen it.

B Abandon the attribution requirement and report accuracy metrics only

Drops a stated model risk requirement because one service is unavailable, when the underlying capability is entirely available.

C Use a foundation model to write explanations of each declined application

Produces a plausible narrative, not an attribution. Module 12's warning: a stated reason is not evidence of the computation.

D Compute the metrics directly with pandas and SHAP in Pipelines, logging to a SageMaker AI MLflow App

The documented replacement path. The bias formulas are published and SHAP is the engine Clarify itself uses; MLflow gives the audit trail.

THE SIGNAL

Clarify is a packaging of public mathematics. Losing the service does not lose the capability, which is why this replacement is a good one.

> Four data scientists have spent a quarter improving a forecasting model, working in separate notebooks and running training jobs ad hoc. Results were compared in a shared spreadsheet.
>
> The best model from eight weeks ago outperforms anything since, and nobody can reproduce it. The spreadsheet records its score but not its hyperparameters or which dataset version it used.

What should they put in place?

A SageMaker managed MLflow, tracking runs from notebooks, training jobs and pipelines

B SageMaker Debugger, to capture tensors during each training run

C A stricter spreadsheet template with mandatory fields

D Clarify processing jobs after each run, to record feature attributions

◆ PAUSE AND DECIDE

What should they put in place?

A SageMaker managed MLflow, tracking runs from notebooks, training jobs and pipelines

Captures parameters, metrics, artefacts and data version automatically, from every place the team works. Registered models flow into the Model Registry.

B SageMaker Debugger, to capture tensors during each training run

Diagnoses a run in flight. It does not organise or compare runs afterwards, and it is closed to new customers.

C A stricter spreadsheet template with mandatory fields

Manual recording failed once and will fail again. It also cannot capture the container image or dataset version reliably.

D Clarify processing jobs after each run, to record feature attributions

Explains why a model predicts as it does. It does not record which configuration produced it, which is the actual gap.

THE SIGNAL

The tracking has to be automatic and it has to reach every place people work. That combination is what rules out the spreadsheet.

A lending team reviews their model's global feature importance and finds that 'days_since_account_opened' contributes far more than any other feature, several times more than income or payment history.

The model scores 0.96 AUC on validation. A colleague suggests removing the feature to make the importance profile look more balanced.

How should you respond?

A Agree. A dominant feature makes the model fragile, so remove it and retrain

B Disagree. High importance means the feature is highly predictive, which is desirable

C Investigate whether the feature is leaking, before deciding anything about it

D Apply L1 regularisation so the model relies on it less

◆ PAUSE AND DECIDE

How should you respond?

A. Agree. A dominant feature makes the model fragile, so remove it and retrain

Removes the symptom without understanding it. If the feature is legitimate you lose signal; if it is leaking, you never learn why.

B. Disagree. High importance means the feature is highly predictive, which is desirable

Predictive of what, and available when? A leaked feature is extremely predictive in validation and useless at prediction time.

C. Investigate whether the feature is leaking, before deciding anything about it

Dominance plus 0.96 AUC is the leakage signature from module 05. Ask whether the value exists, with this meaning, at prediction time.

D. Apply L1 regularisation so the model relies on it less

Regularisation constrains overfitting to noise. A leaked feature is not noise, so the model is right to weight it heavily.

THE SIGNAL

This is what explainability is for. The metric said the model was excellent; the attribution said look more closely.

What to take from this module

- **SHAP is local or global. PDP is always global.** Match the scope to the audience.
- **Online explainability** when the explanation must arrive with the prediction, at a latency cost.
- **Post-training bias needs predictions**, so it needs a model or endpoint, and it is slower.
- **Clarify is closed, and the mathematics and SHAP are both public.** Compute and log to MLflow.
- **Track experiments automatically**, or in six months you will not be able to reproduce anything.

> ON THE EXAM
>
> That closes Domain 2, at 24 percent of the exam. Module 20 opens Domain 3 with choosing an inference option.